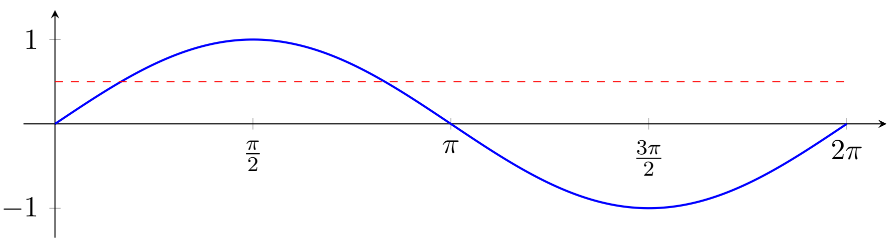
Solving Trigonometric Equations on Restricted Intervals

Precalculus Trigonometric Identities and Equations

Linear trigonometric equations, identity-first rewrites, quadratics in a trigonometric function, and periodic models — all on restricted intervals

Why the interval is part of the question. A trigonometric equation has infinitely many solutions, spaced one period apart. Restricting the interval is what turns it into a question with a finite, listable answer. The graph below is the whole idea: $y = \sin x$ meets the line $y = \frac{1}{2}$ twice on $[0, 2\pi)$ — so "solve $2\sin x - 1 = 0$ on $[0, 2\pi)$" has exactly two answers, and reporting one of them is only half an answer.



Directions. Give *every* solution in the stated interval, in exact form (radians unless the problem says degrees). Show the rewrite or factoring that reduced the equation, and state the units where the problem has them.

1. Solve on $[0, 2\pi)$: $2\sin x - 1 = 0$

Use the graph above to predict how many solutions you should find before you compute them.

Answer: ________________

2. Solve on $[0, 2\pi)$: $2\cos x + \sqrt{3} = 0$

Answer: ________________

3. Solve on $[-\pi, \pi)$: $\tan x + 1 = 0$

The period of tangent is π, not 2π, and the interval is not the usual one — both facts change the count.

Answer: ________________

4. Verify the double-angle identity $\cos 2x = 1 - 2\sin^2 x$ by starting from $\cos 2x = \cos^2 x - \sin^2 x$ and using $\cos^2 x + \sin^2 x = 1$.

Write the identity you end with — you will use it in problem 5.

Answer: ________________

5. Solve on $[0, 2\pi)$: $\cos 2x + 3\sin x = 2$

Two different functions of two different angles cannot be combined. Use problem 4's identity to write everything in terms of $\sin x$ first.

Answer: ________________

6. Solve on $[0, 2\pi)$: $2\cos^2 x - \cos x - 1 = 0$

This is a quadratic in $\cos x$. Factor it, then solve each factor separately — and check whether each value of $\cos x$ is attainable.

Answer: ________________

7. A Ferris wheel turns at a constant rate. A rider's height above the ground, in metres, t seconds after boarding, is

$$h(t) = 9 - 8\cos\left(\frac{\pi t}{15}\right).$$

Find the rider's height 10 seconds after boarding.

Answer: ________________ m

8. For the same Ferris wheel, $h(t) = 9 - 8\cos\left(\frac{\pi t}{15}\right)$ metres. Find every time t in the interval $[0, 30)$ seconds at which the rider is exactly 13 metres above the ground.
The interval is one full turn of the wheel. Give both times.

Answer: ________________ s

9. Solve on $[0, 2\pi)$: $\sin 2x = \cos x$
Rewrite $\sin 2x$, move everything to one side and factor. Dividing both sides by $\cos x$ would destroy solutions — do not do it.

Answer: ________________

10. Solve on $[0^\circ, 360^\circ)$: $2\sin^2\theta - \sin\theta - 1 = 0$

Degrees this time. Factor the quadratic in $\sin\theta$, solve each factor, and give all solutions in degrees.

Answer: ________________